



STRUCTURAL CHANGE THROUGH STRUCTURAL DEPENDENCIES: A CORE-PERIPHERY MODEL OF SUFFICIENCY TRANSITIONS

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1 • Context

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What does a reduction of throughput in the Core do to the Periphery?

- **Unequal exchange.** Core countries sustain their consumption through a net appropriation of labour and resources via unequal exchange in international trade. (Dorninger et al., 2021; Hickel et al., 2022, 2024)
- **The twin problem of global dependencies.** (Grabner & Radkowsch, 2023) The structures that let the Core externalise its social and ecological costs are the very structures the Periphery now depends on: physically, monetarily, financially. **The argument for sufficiency is also the obstacle to it.**

Channels of dependency

Physical & technological Production requires imported inputs in near-fixed proportions.

Monetary & external constraint The Core settles in the money it issues, the Periphery has to earn it, so only the Periphery's balance of payments binds. (Alami et al., 2023; Althouse, 2025; Olk et al., 2023)

Terms of trade Wage differentials and specialisation asymmetries set the price at which the Periphery sells. (Amin, 1976; Emmanuel, 1972; Lewis, 1954; Prebisch, 1950; Ricci, 2022)

Property income Profits earned in the Periphery accrue to Core owners. (Parnreiter, 2024)

Financial fragility Debt service and exposure to capital-flow reversals tighten as vulnerability rises. (Althouse, 2025; Songwe, 2024)

What the literature already shows

Fevereiro et al. (2025): a uniform material-footprint cut in the Core, in MRIO: output, employment and the trade balance all deteriorate in the Periphery.

Grabner and Radkowsch (2026): measures the exposure itself: how much Periphery activity runs through import, financial and technological dependence on the Core.

Nag (2026): the Great Recession as a proxy Core contraction: falling import prices and a shifting trade structure leave parts of the Periphery better off.

Althouse et al. (2020): a Core-Periphery coordination game: only Core degrowth combined with green industrial change and investment in the Periphery stays within limits.

Leoni et al. (2025): two-region SFC: a unilateral post-growth transition in the Core triggers a balance-of-payments crisis in the Periphery. Cooperation averts it.

2 • Model: three changes of closure

BRICK 0 • LEONTIEF

Closure. $\mathbf{f} = \mathbf{f}^C + \mathbf{f}^P$: Periphery final demand stays exogenous.

The physical dependencies of production are the fixed coefficients $\mathbf{A}$, through which a Core shock propagates to the Periphery.

- Core final demand is shocked by $\mathbf{d}$, carrying both level and composition
- We read off Periphery retained income, employment, absorption, trade balance and footprint
- No absorption: the Periphery can only shrink.

BRICK 1 • PERIPHERY RESPENDING

Closure. A share c of retained income is respent in the Periphery on a basket α , on top of a fixed autonomous component.

- Income harm is **scaled up** by a multiplier: **the MRIO number is a lower bound.** The closure does not re-rank cuts by income harm per tonne, but it does re-rank them by employment
- A Core cut always tightens the Periphery's external position.

BRICK 2 • MONETARY CONSTRAINT

Closure. The spending rule is capped by the foreign exchange actually available $\Phi = X - M + F$.

Financing capacity F represents the capacity to run a trade deficit out of reserves, net of a debt service fixed in foreign currency.

At fixed exchange rates, adjustment falls on **quantities**: imported intermediates are rationed, output is input-capped.

A non-linearity: demand-determined when slack, forex-determined when binding.

BRICK 3 • REABSORPTION

Closure. Labour released by the shock is partly reabsorbed on a basket χ .

- **Slack:** reabsorption is free
- **Binding:** reabsorption competes with household consumption for foreign exchange, so full reabsorption requires a channel that is forex-neutral or forex-saving

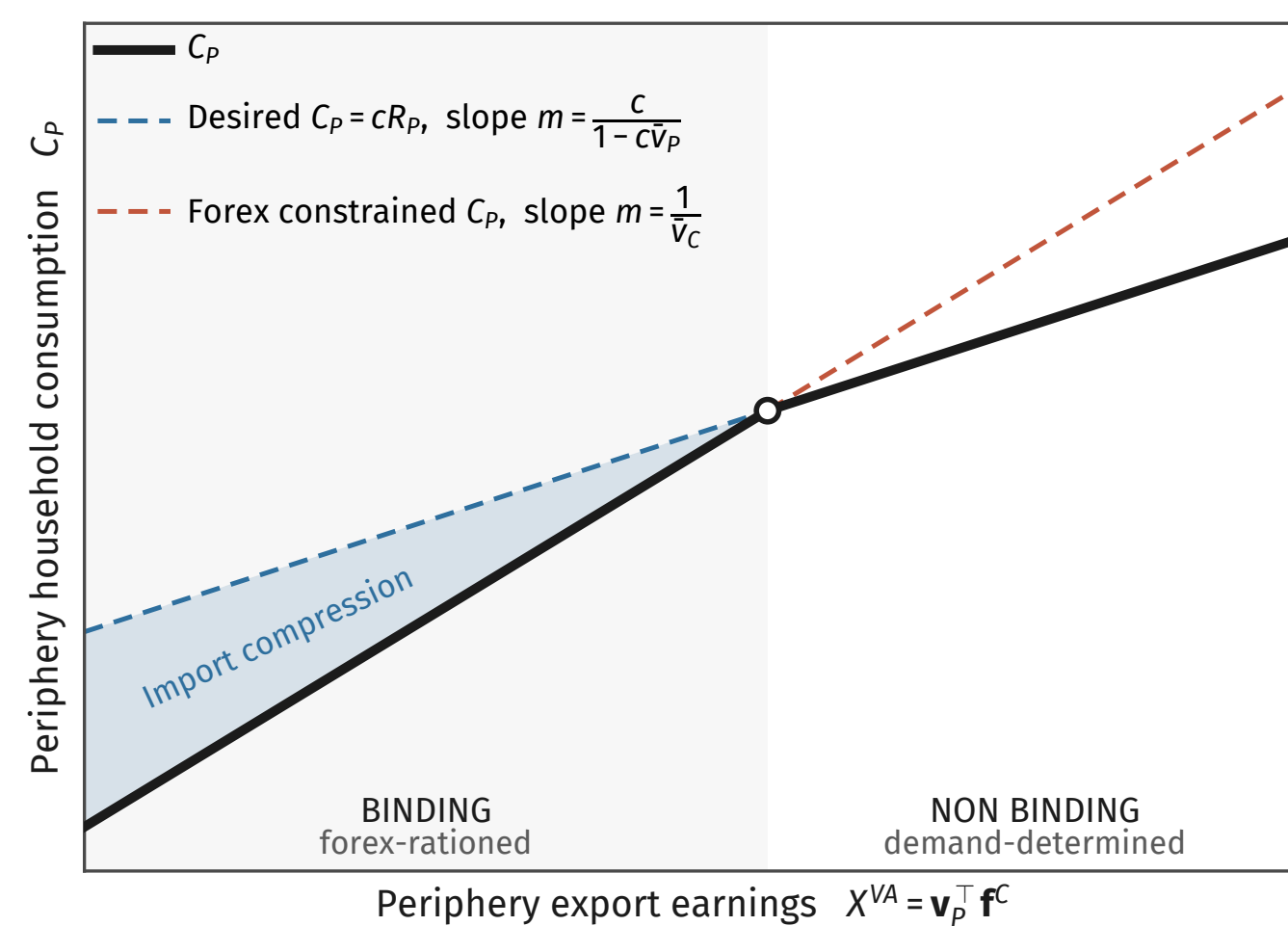


Figure 1. Brick 2, the regime switch. Consumption follows desired spending until foreign exchange runs out, then follows what exports can pay for.

Empirics

EXIOBASE 3.10.2, product-by-product, year 2022.

- The cut and the Core / Periphery split of Fevereiro et al. (2025)
- Financing capacity F : three-months-of-imports reserve floor, reserves above it drawn down evenly over one year.

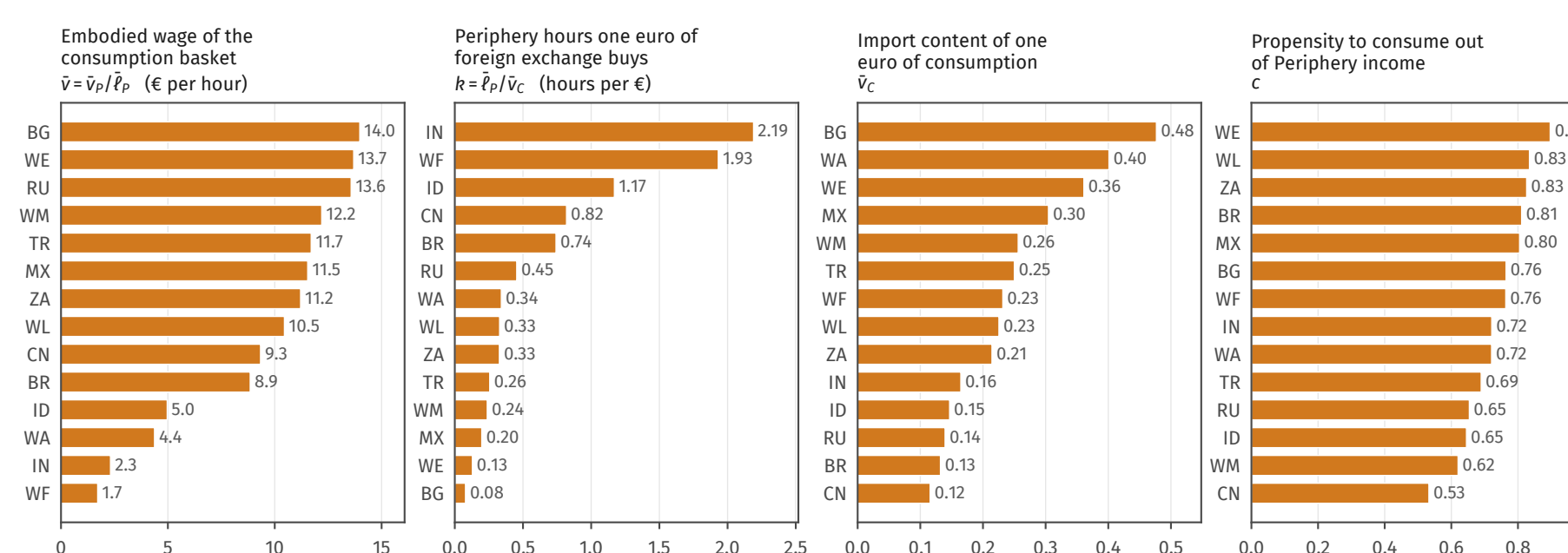


Figure 2. Characteristics per region

Summary of the algebra

Brick	Closure of periphery demand	Feedback consumption C_P	Multiplier m	Output response $d(\bar{\mathbf{z}}^T \mathbf{f})$
0	$\bar{\mathbf{f}}^P$	0	0	$\mathbf{z}^T \mathbf{d}$
1	$C_P \frac{\alpha}{\beta} + \bar{\mathbf{A}} \frac{\beta}{\beta}$	$c R_P$	$\frac{c}{1 - c \bar{v}_P}$	$(\mathbf{z}^T + m \bar{\mathbf{z}} \bar{\mathbf{v}}_P^T) \mathbf{d}$
2	$C_P \frac{\alpha}{\beta} + \bar{\mathbf{A}} \frac{\beta}{\beta}$	$\min \left\{ c R_P, \frac{X^{VA} + F - \bar{v}_C^{\beta} \bar{\mathbf{A}}}{\bar{v}_C} \right\}$	$\frac{c}{1 - c \bar{v}_P}, \frac{1}{\bar{v}_C}$	$(\mathbf{z}^T + m \bar{\mathbf{z}} \bar{\mathbf{v}}_P^T) \mathbf{d}$
3	$C_P \frac{\alpha}{\beta} + \bar{\mathbf{A}} \frac{\beta}{\beta} + a \chi$	$\min \left\{ c R_P, \frac{X^{VA} + F - \bar{v}_C^{\beta} \bar{\mathbf{A}} - a \Phi_{\chi}}{\bar{v}_C} \right\}$	$\frac{c}{1 - c \bar{v}_P}, \frac{1}{\bar{v}_C}$	$(\mathbf{z}^T + m \bar{\mathbf{z}} \bar{\mathbf{v}}_P^T) \mathbf{d} + \left(\mathbf{z}^T \chi + \bar{\mathbf{z}} \frac{\partial C_P}{\partial a} \right) \mathbf{d} a$ the cut the reallocation

Key notation. $\bar{\mathbf{z}}_P$ intensity z per euro of Periphery consumption $\bar{\mathbf{A}}, \beta$ autonomous demand, and its basket R_P, C_P Periphery retained income and the part respent X^{VA}, F export earnings in value added and net financing capacity Φ_{χ} net forex cost of basket χ , per hour

3 • Results

What does the behavioural response change?

- The income loop **rescales** income harm per tonne but leaves the ranking of the most affected sectors intact.
- Brick 0 implicitly assumes **unlimited foreign exchange**. The income loop lets part of the loss be taken by the trade balance rather than by consumption. With a binding external constraint, the whole shock becomes a domestic contraction.
- **Rebound appears only with reabsorption**, and only in world material use, never in the Core's own footprint: it bites when the Core cuts a labour-heavy, material-light Periphery export.

What constrains how much labour can be reabsorbed?

Characteristic of the Periphery	Object	Outcome
High import content of consumption	$\bar{v}_C$	fewer hours per euro of forex diverted $\rightarrow$ easier
High labour intensity of consumption	$\bar{\ell}_P$	forex diverted is hour-expensive $\rightarrow$ harder
High reserves net of debt service	F	larger forex buffer $\rightarrow$ easier
High propensity to consume out of income	c	closer to the constraint $\rightarrow$ harder

Designing reabsorption

When **non-binding**, maximise income per hour and let the loop do the rest.
When **binding**, only forex-saving channels raise the capacity to fund everything else.

Relocalisation and **South-South trade** (can allow full absorption) > **Subsistence** (forex neutral) > **Public provisioning**

Designing sufficiency

1. Cut sectors by the **material carried per Periphery hour**, not by Periphery value added per tonne: only the first survives the behavioural response. Labour-heavy, material-light Periphery exports look like cheap cuts in MRIO accounting and are the worst cuts once the Periphery reabsorbs.
2. Large cuts can flip the Periphery into the binding regime, so **pacing matters**.

4 • Limitations

- Static: no investment, no debt dynamics
- Fixed prices, no endogenous terms of trade
- Financing capacity F exogenous
- Fixed technical coefficients, no input substitution
- No distribution inside the Periphery; spending propensity and baskets held constant under the shock

5 • Take-away

- A stylised **two-region model** propagating a compositional sufficiency shock through physical dependencies.
- Three **changes of closure** (responding, the external monetary constraint, reabsorption of released labour) turn a mechanical contraction into a behavioural response with a **regime switch**.
- On EXIOBASE, the closure changes the size and the ranking of the harm, and yields concrete **cut-design and instrument** recommendations.

